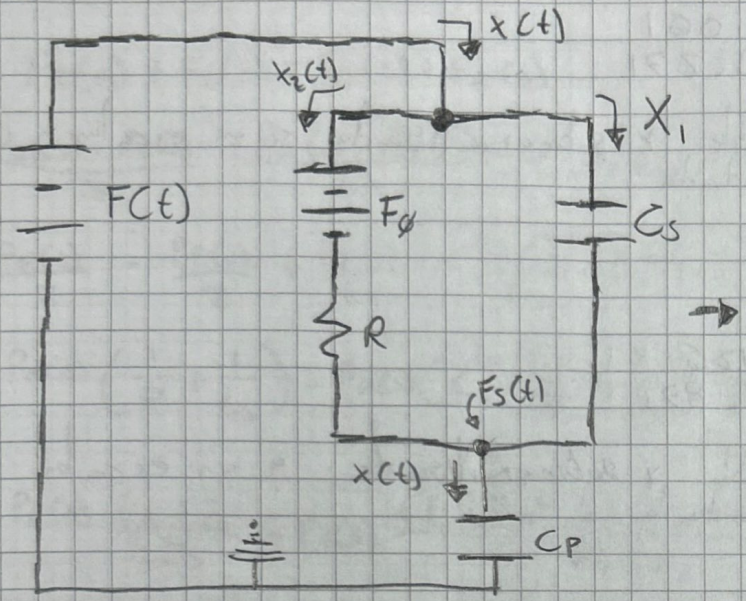
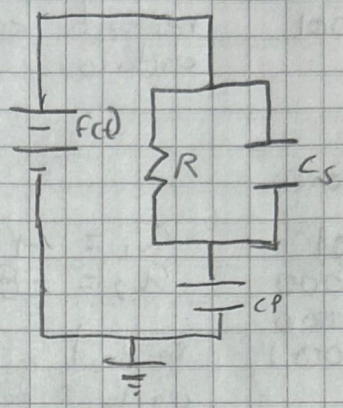
Practica: Modelado y control del sistema musculoesqueletico



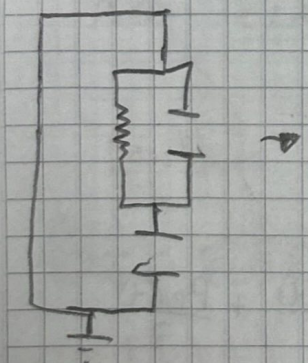
(Quitando F_ϕ)

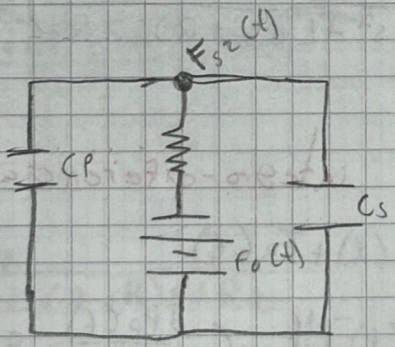


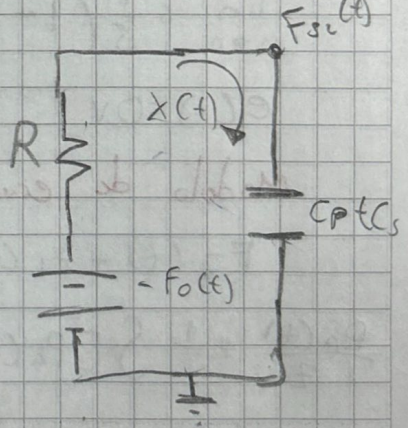
$x(t) = x_1(t) + x_2(t)$

$f_S(t) = F_{S_1}(t) + F_{S_2}(t)$

(Quitando F(t))







Análisis matemático

Debido a que tiene dos fuentes se realiza por superposición

Circuito 1: Apagado $F_0(t)$

$$X(t) = X_1(t) + X_2(t)$$

$$X(t) = C_p \frac{dF_{s_1}(t)}{dt}$$

$$X_1(t) = C_s \frac{d[F(t) - F_{s_1}(t)]}{dt}$$

$$X_2(t) = \frac{F(t) - F_{s_1}(t)}{R}$$

$$C_p \frac{dF_{s_1}(t)}{dt} = C_s \frac{d[F(t) - F_{s_1}(t)]}{dt} + \frac{F(t) - F_{s_1}(t)}{R}$$

$$C_p s F_{s_1}(s) = C_s s [F(s) - F_{s_1}(s)] + \frac{F(s) - F_{s_1}(s)}{R}$$

$$C_p s F_{s_1}(s) + C_s s F_{s_1}(s) + \frac{F_{s_1}(s)}{R} = C_s s F(s) + \frac{F(s)}{R}$$

$$\left(C_p s + C_s s + \frac{1}{R}\right) F_{s_1}(s) = \left(C_s s + \frac{1}{R}\right) F(s)$$

$$\frac{R(C_p + C_s)s + 1}{R} F_s(s) = \frac{C_s R s + 1}{R} F(s)$$

$$F_{s_1}(s) = \frac{\frac{C_s R s + 1}{R}}{\frac{R(C_p + C_s)s + 1}{R}} F(s)$$

$$F_{s_1}(s) = \frac{C_s R s + 1}{R(C_p + C_s)}$$

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Circuito 2: Apagado F(t)

$$-aF(t) = Rx(t) + \frac{1}{(C_P+C_S)}\int x(t)dt$$

$$F_{S2}(t) = \frac{1}{(C_P+C_S)}\int x(t)\,dt$$

$$-aF(s) = Rx(s) + \frac{x(s)}{(C_P+C_S)s}$$

$$F_{S2}(s) = \frac{x(s)}{(C_P+C_S)s}$$

$$F(s) = -\frac{1}{a}\left(R + \frac{1}{(C_P+C_S)s}\right)x(s) = -\frac{1}{a}\,\frac{R(C_P+C_S)s+1}{(C_P+C_S)s}\,x(s)$$

$$\frac{F_{S2}(s)}{F(t)} = \frac{\frac{x(s)}{(C_P+C_S)s}}{\frac{1}{a}\,\frac{R(C_P+C_S)s+1}{(C_P+C_S)s}\,x(s)} = -\frac{a}{R(C_P+C_S)s+1}$$

$$F_{S2}(s) = -\frac{aF(s)}{R(C_P+C_S)s+1}$$

Función de transferencia

$$F_S(s) = F_{S_1}(s) + F_{S_2}(s) = \frac{C_S R_S + 1}{R(C_P + C_S)s + 1} F(s) - \frac{q F(s)}{R(C_P + C_S)s + 1}$$

$$F_S(s) = \frac{C_S R_S + 1 - q}{R(C_P + C_S)s + 1} F(s)$$

$$\frac{F_S(s)}{F(s)} = \frac{C_S R_S + 1 - q}{R(C_P + C_S)s + 1}$$

Error en estado estacionario

$$e(s) = \lim_{s \to 0} s F_S \left(1 - \frac{F_S(s)}{F(s)}\right) = \lim_{s \to 0} s \frac{1}{s} \left(1 \frac{RC_S s + 1 - q}{(RC_P + RC_S)s + 1}\right) = q$$

Estabilidad lazo abierto

$$(RC_P + RC_S)s + 1 = 0$$

$$\lambda = -\frac{1}{RC_P + RC_S}$$

El sistema es estable